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基于云联合同态加密的隐私增强联合学习

Privacy-Enhanced Federated Learning Through Homomorphic Encryption With Cloud Federation

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***Abstract*—With the exponential growth of data at edge environments, federated learning has emerged. Nevertheless, it encounters persistent challenges in defending against malicious attacks, such as inference attacks and reconstruction attacks. In this paper, we enhance the security of federated learning by integrating cryptographic primitives, including Elliptic Curve Cryptography (ECC) and homomorphic encryption, with cloud federation to protect private user data from being inferred by adversaries using localized models. Specifically, we employ two cloud service providers (CSPs) to manage encrypted model aggregation and global model decryption services separately, which ensures robust privacy protection and provides a strong solution for safeguarding sensitive information. ECC is utilized to encrypt the private key used in homomorphic encryption. Additionally, we partition the ECC private key into multiple sub-secrets and distribute them among the clients to defend against collusion attacks during the decryption process of the global model. Furthermore, we harness the gradient offset between local and global models to design an optimized objective function tailored for clients handling non-Independent and Identically Distributed (non-IID) and unbalanced distributed datasets. Our experimental results show that our method improves accuracy by 1.5% over the baseline, while also ensuring robust privacy protection.**

***Index Terms*—Cloud Federation, Federated Learning, Homomorphic Encryption, Elliptic Curve Cryptography**

I. Introduction

With the advent of the Internet of Things (IoT) [1], smart devices in our surroundings continuously generate substantial amounts of data. Centralized cloud computing is increasingly inadequate for meeting diverse needs. The emergence of distributed computing, exemplified by technologies like cloud federation [2] and federated learning, addresses the complexity and diversity of needs that exceed the capabilities of a single cloud service provider. Nevertheless, in a federated learning system, if a client's updated model is transmitted to the central server without adequate protection, it becomes vulnerable to compromising the user's data privacy by observing the client's model parameters and gradient variations. To address the above problem, certain researchers have proposed federated learning schemes that utilizes differential privacy to protect the security of client updated models [3][4][5].

Unfortunately, introducing noise to local models through differential privacy for protection can degrade the performance when training models. To safeguard the privacy of users' sensitive data without compromising the training model's performance, we propose a federated learning scheme based on homomorphic encryption [6] and Elliptic Curve Cryptography (ECC) [7] within a cloud federation framework to improve data privacy protection and distributed computing efficiency. Firstly, we utilize two CSPs: the Cloud Server (CS) manages model aggregation in ciphertext, while the Auxiliary Server (AS) reconstructs the private key for model decryption. Secondly, we employ homomorphic encryption to encrypt model parameters and utilize ECC to encrypt the private key of homomorphic encryption. To thwart collusion attack, we distribute the subkeys generated by the ECC private key to the clients. Then, each client uploads both the model to the CS and the subkey to the AS simultaneously. Additionally, we design an optimized objective function for non-IID distributed datasets by introducing the gradient variations between the global and local models to alter the gradient descent direction during local model training. In summary, the main contributions of this article are as follows:

1) We present a novel privacy-preserving federated learning approach that incorporates cloud federation and homomorphic encryption to address security and privacy issues. This allows for the separation of homomorphic aggregation computation and the global model decryption, which ensures none of the CSPs has direct access to the client's precise model.
2) The objective function is optimized with the gradient difference between local and global models to achieve superior efficiency compared to other common federated learning algorithms with privacy-preserving techniques.
3) We also propose a novel secret key distribution scheme where homomorphic encryption is to encrypt the model parameters, and ECC encrypts the private key of homomorphic encryption. Furthermore, we segment the ECC private key into multiple subkeys distributed among clients. This approach is designed to mitigate collusion risks between clients and CSPs during the training process.
4) We conduct comprehensive experiments based on real-

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world datasets to validate the performance of the proposed federated learning scheme.

II. RELATED WORK

Federated learning is a machine learning approach that trains models by sharing model updates among distributed data sources, without centralizing the original datasets. It enables model training in a privacy-preserving, efficient, and adaptable manner by sharing model updates among distributed clients. Li et al. introduced Ditto, a personalized federated learning framework that addresses robustness and fairness across devices in statistically heterogeneous networks [8]. Karimireddy et al. identified federated averaging's limitations with heterogeneous data and proposed SCAFFOLD, which reduces client-drift using control variates, improving convergence rates and communication efficiency [9]. While federated learning enhances user privacy by enabling model training across multiple devices or data sources without centralizing the original dataset, it remains vulnerable to certain privacy attacks. Bhowmick et al. elaborated on the reconstruction attack and introduce a countermeasure: localized differential privacy [10]. Nasr et al. and Shokri et al. exposed significant privacy leakage in federated learning models using white-box inference attacks on stochastic gradient descent algorithms [11]. To counter these attacks, various privacy-preserving techniques are typically used in federated learning to enhance privacy protection. Among these, differential privacy [12] techniques and encryption methods are the most commonly employed.

Wu et al. introduced an adaptive federated learning scheme with adaptive gradient descent and differential privacy, enhancing multi party collaborative modeling efficiency and privacy robustness [4]. Hu et al. proposed a privacy-preserving approach for personalized federated learning using differential privacy on distributed user data, addressing user heterogeneity [3]. Wu et al. introduced FedML-HE, which uses homomorphic encryption to selectively encrypt sensitive parameters, reducing overhead and providing customizable privacy protection in large models [13]. Wang et al. introduced PPFLHE, a privacy-preserving federated learning scheme for healthcare data using homomorphic encryption and access control, with an ack mechanism to manage latency and communication [14]. Zhang et al. integrated federated learning with cryptographic protection in an IoT healthcare system, prioritizing encrypted dataset quality for privacy and training resilience [15].

III. PRELIMINARIES

~~A. Federated learning~~

In federated learning, each client trains a model using local data and sends the trained model to the central server. The central server aggregates these local models to create a global model, which is then distributed back to all clients for updates. The standard federated learning architecture follows the client-server model, comprising multiple clients denoted as $C = \{C_1, C_2, ..., C_K\}$ and a central server S, where C_i holds the data D_i ($D = \{D_1, D_2, D_3, ..., D_i, ..., D_K\}$). The well-known federated average model (FedAvg) [16] was conceived within the client-server model framework. The training process in federated learning from round t to round $t+1$ can be represented by Eq. 1 and 2.

$$\mathcal{L}_f = \frac{1}{|D_i|} \sum_{<x_j, y_j> \in D_i} \left\| L(x_j, w_i^t) - y_j \right\|_2 \tag{1}$$

$$w_i^{t+1} = w_i^t - \eta_t \mathcal{L}_f,\ w^{t+1} = \frac{n_k}{n} \sum_{j=1}^{K} w_j^{t+1}, \tag{2}$$

where $\mathcal{L}_f$ represents the loss function, w denotes the global model weight vector, w_i^t represents the client c_i's local model weight vector, and η_t is the learning efficiency in round t. Furthermore, federated learning can be formulated as an optimization problem aimed at minimizing the loss function, shown in Eq. 3.

$$\begin{aligned} &\min_{w \in \mathbb{R}^d} f(w) = \frac{n_k}{n} \sum_{j=1}^{K} F_j(w), \\ &s.t.\, F_j(w) = \frac{1}{n_j} \sum_{z \in D_j} f_z(w), \end{aligned} \tag{3}$$

where $F_j(w)$ denotes the local empirical objective and $f(w)$ denotes the global empirical objective.

~~B. Fully homomorphic encryption~~

Fully Homomorphic Encryption (FHE) [17] is a crucial algorithm for secure computing, enabling third parties to perform addition, subtraction, multiplication, and division operations on encrypted data without knowing the inputs or the results of the computation. In this paper, we utilize the CKKS [18] homomorphic encryption scheme to safeguard data security and privacy. The security of the CKKS homomorphic encryption scheme builds on the complexity of the LWE problem [19]. The difficulty of the LWE problem is intricately tied to the challenges of solving the SVP [20] and CVP [21].

The CKKS algorithm consists of six components: key generation, encoding, decoding. encryption, decryption and computation. The key generation section is shown below:

$\mathbf{CKKS.GenKey}(\lambda)$: Generate the private key (sk), public key (pk), and evaluation key (evk) according to the specified security parameters λ.

$\mathbf{CKKS.Ecd}(g)$: For a vector g, convert it to a polynomial $m(x)$.

$\mathbf{CKKS.Dce}(m(x))$: For a polynomial $m(x)$, convert it to a vector g.

$\mathbf{CKKS.Enc}(m(x), pk)$: For a polynomial $m(x)$, output $c = (b, a) \in \mathbb{R}_q^2$.

$\mathbf{CKKS.Dec}(c, sk)$: For a ciphertext $c = (b, a) \in \mathbb{R}_q^2$ output $m(x)$.

$\mathbf{CKKS.ADD}(c_0, c_1)$: $c_r \leftarrow c_0 + c_1 mod \quad p^l$.

$\mathbf{CKKS.Mul}(c_0, c_1)$: $c_r \leftarrow c_0 \times c_1 mod \quad p^l$.

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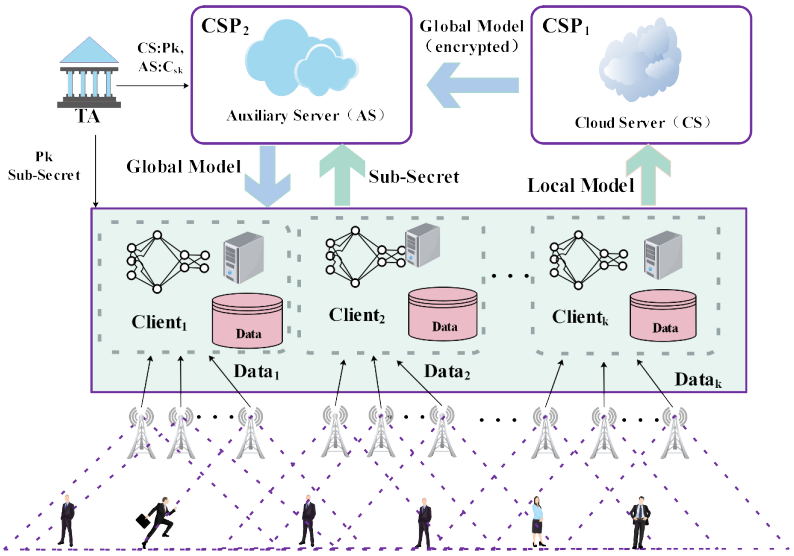


Fig. 1: System Model

IV. System Model

As shown in the Fig. 1, the system model consists of five entities: Cloud server(CS), Auxiliary Server(AS), Client, User and Trusted Authority(TA). We designate CSP_1 as the CS and CSP_2 as the AS. They collaborate to complete the entire model aggregation process. Since they possess different secret keys and access distinct data, neither CSP can obtain the model parameter information uploaded by the clients during the system's execution. These ensures the overall security and reliability of the system. The TA plays a crucial role in securely distributing secret keys to guarantee their confidentiality, integrity, and trustworthiness.

A. ~~Secret sharing protocols based on ECC and Shamir's thresholds~~

Shamir's threshold secret sharing scheme [22] is a cryptographic technique based on polynomial interpolation. It is designed to distribute a secret among multiple participants in a way that ensures the original secret can only be reconstructed if a sufficient number of participants collaborate. The core idea of this scheme is to use a polynomial function to divide the secret. In this paper, we suppose that the base point of the ECC is E. We use the public key Q in ECC to encrypt the private key(sk) in the CKKS scheme. Additionally, we employ shamir's threshold secret sharing scheme to divide the ECC private key d into multiple sub-secrets.

We determine the number of clients, K, to be the number of holders of these sub-secrets, and select z as the threshold value required to reconstruct the private key. The ECC private key is then divided into K sub-secrets using shamir's threshold secret sharing protocol, with the threshold set to t. This enables the reconstruction of the ECC private key when z clients collaborate. We select K distinct values of x and apply them to the polynomial $H(x)$, resulting in K pairs $(H(x), x)$. These pairs are then distributed among K clients, with each client responsible for safeguarding their respective pair $(H(x), x)$ as a secret.

$$C_{sk} = \{C_1, C_2\} = \{r \cdot E, sk + rQ\} \tag{4}$$

$$H(x) = d + b_1 \cdot x + b_2 \cdot x^2 + ... + b_{z-1} \cdot x^{z-1} \tag{5}$$

$$\begin{cases} H(x_1) = d + b_1 \cdot x_1 + b_2 \cdot x_1^2 + ... + b_{z-1} \cdot x_1^{z-1} \\ H(x_2) = d + b_1 \cdot x_2 + b_2 \cdot x_2^2 + ... + b_{z-1} \cdot x_2^{z-1} \\ H(x_3) = d + b_1 \cdot x_3 + b_2 \cdot x_3^2 + ... + b_{z-1} \cdot x_3^{z-1} \\ \qquad\qquad ... \\ H(x_K) = d + b_1 \cdot x_n + b_2 \cdot x_n^2 + ... + b_{z-1} \cdot x_K^{z-1} \end{cases} \tag{6}$$

The reconstruction process of the ECC private key d is facilitated through the Laplace difference method. After the reconstruction is finalized, the ECC private key d can be retrieved. Utilizing this ECC private key d, sk can be obtained by decrypting C_{sk}. The process is outlined in Eq. 7 and Eq. 8:

$$H_i(x) = \prod_{j \neq i}^{1 \leq j \leq K} \frac{(x - x_j)}{x_i - x_j}, \; H(x) = \sum_{i=1}^{K} H(x_i) H_i(x) \tag{7}$$

$$sk = C_2 - d \cdot C_1 = sk + r(dE) - d(rE) \tag{8}$$

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B. Aggregation model

The objective of federated learning is to fit a single global model over all local data in the network, denoted as w. The aim is to solve the following optimization problem:

$$\min_{w} G(w) = Aggregate(F_1(w), F_2(w), ..., F_k(w)), \quad (9)$$

where $F_i(w)$ is a local target of client c_i, $G(\cdot)$ is the global target of the federated learning and $Aggregate(\cdot)$ is a function that aggregates the local targets F_i from each client.

The dataset used for training on each client may have different distribution. Since the training model of client c_i only considers using its own dataset D_i, and does not account for the global dataset, the local models may not fully capture the global data distribution, leading to significant differences from the global models. To address this, we optimize by considering the offset between the gradient of the client's local model and the gradient of the global model. This adjustment helps the local models align more closely with the global model during aggregation. Assuming that $G^t(w)$ and $G^{t+1}(w)$ represent the global model at rounds t and $t+1$ respectively, g_g^t represents the global gradient at round t, and g_i^t represents the local gradient of the i client at round t. The global gradient g_g^t and local gradient g_i^t are computed as shown in Eq.10:

$$g_g^t = \frac{\nabla G^t(w) - \nabla G^{t+1}(w)}{\eta_t}, \; g_i^t = \frac{\nabla F_i^t(w) - \nabla F_i^{t+1}(w)}{\eta_t} \quad (10)$$

To optimize the local model efficiently, it's important to calculate the distance between the gradients at rounds g_g^t and g_i^t. Additionally, introducing a scaling factor σ allows for more precise control over the offset of the local gradient, further enhancing the optimization process. Consider the offset between the local model and the global model. The optimization model for each client is given by Eq. 11:

$$\min_{w_i} \mathcal{L}_i(w_i, w) = F_i(w) + \frac{||g_i^t - g_g^t||_2}{2\eta_t \sigma} ||w_i - w||_2 \quad (11)$$

$$\tau = \frac{||g_i^t - g_g^t||_2}{\eta_t \sigma} \quad (12)$$

For clarity, we denote the offset between the local model and the global model as τ. The τ controls the interpolation between the local and global models.

C. Secure aggregation process

In our model, we can provide personalized detection for individual users while preserving the characteristics of the global data. In addition, we ensure the user's privacy is protected during the model aggregation process.

The model described in our paper is detailed in Fig. 2. Before starting the federated learning model, the server verifies the participating clients and initializes the training

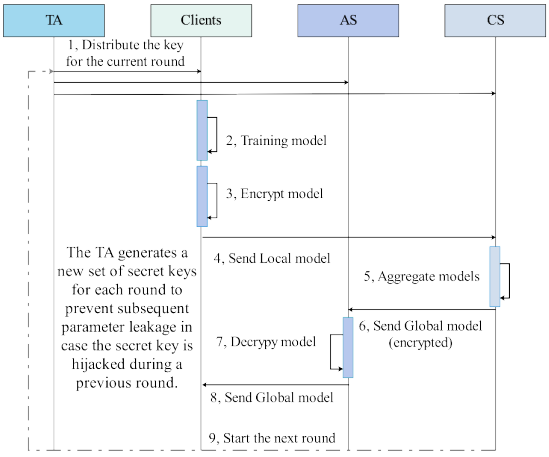


Fig. 2: Secure Aggregation Process

models, including initial learning rate η, maximum iterations T for convergence, initialized global model w, and shamir's threshold secret sharing threshold z. At the same time, the security parameters and pairwise keys are then generated by TA to ensure privacy and security throughout training. The number of generated key pairs equals the number of training rounds. After informing all clients about the training network, parameters, and initialized global model, initialization concludes. Each client receives a secret share of d and pk. This setup enables a privacy-preserving federated learning process. Clients train their models using local data, uploading encrypted parameters to the server. The server aggregates these local models to generate the globle modle.

The secure aggregation process is outlined in Algorithm 1. Each client trains its local model W using its own data, iterating over parameter updates H times within each round. After local training, clients encrypt their model parameters W_i^t into ciphertext C_i^t to prevent interception by external adversaries. Once the server collects encrypted models from at least z clients, aggregation begins. The server assigns weights based on client data usage and completes parameter aggregation in ciphertext. Next, the AS reconstructs ECC private key d from z sub-secrets uploaded by clients. Using d, AS decrypts C_{sk} to obtain homomorphically encrypted private key sk. Finally, the AS use sk decrypts C^{t+1} to obtain updated parameters W^{t+1} after one round. After each round, current keys expire, and keys allocated for the next round are utilized.

V. Analysis

A. Security analysis

In this section, we discuss whether the collusion party $\mathcal{A}$ can get the local model w_i^t by breaking the semantic security of homomorphic encryption. Our threat model assumes fewer than z clients are involved in the conspiracy.

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Algorithm 1 Privacy-Preserving Algorithm based on FHE

Input: Dataset from K distributed clients $\{D_1, D_2, ..., D_K\}$, inner local iteration number H, outer global iteration number T, the initial parameters w^0.
for $t = 0$ to $T-1$ **do**
  Server randomly selects a subset of clients S, and sends w^t to them.
  for all clients $s \in S$ in parallel **do**
    for $h = 1$ to H **do**
      $w_s^t = train(w^t, D_s)$
      $\tau = \frac{||g_i^t - g_g^t||_2}{\eta_t \sigma}$
      $w_s^t = w_s^t - \eta_t(\nabla F(w) + \tau(w_s^t - w^t))$
    end for
    $C_s^t \leftarrow enc(w_s^t, pk)$
  end for
  Server aggregates C_s^t
  $D = D_1 + D_2 + ... + D_S$, $C^{t+1} = \frac{D_1}{D} C_1^t + \frac{D_2}{D} C_2^t + ... + \frac{D_e}{D} C_e^t$
  $d \leftarrow recons(e, < x_i, F(x_i) >) i \in S$, $Sk \leftarrow dec(C_{sk}, d)$, $w^{t+1} \leftarrow dec(C^t, Sk)$
  The client activates the keys corresponding to the next round.
end for
Return w^{t+1}

In our system, both the CS and the AS are considered semi-honest, meaning they adhere to the protocol without tampering with transmitted messages or forging public keys and secret shares. Under the semi-honest model, we analyze the security of the scheme. During the encryption process, client c_i encrypts the model parameters and data volume. This encryption is done using the public key pk. The encrypted model parameters are denoted as C_i^t. If the server intends to decrypt the model parameters of client c_i, it requires possession of the homomorphic encryption private key sk. ~~However,~~ sk is encrypted by the TA using elliptic curve cryptography to C_{sk} and stored in the AS. ~~Furthermore~~, should the server attempt to decrypt C_i^t without the homomorphic encryption private key sk, it would need to solve the LWE problem. As of now, this LWE problem is considered NP-hard, rendering it infeasible for the server to solve.

~~Therefore, r~~reconstructing the elliptic cipher private key d to obtain the homomorphic encryption private key sk and decrypt the ciphertext seems to be the only efficient way to achieve this goal. Fortunately, if the adversary $\mathcal{A}$ compromises the AS, the number of colluding clients with $\mathcal{A}$ needs to exceed the threshold z to reconstruct the key. As a result, the CS and clients do not have the ability to decrypt individual local model C_i^t, ensuring that local models C_i^t are protected. The AS will only have access to the aggregated C^t, not the individual C_i^t of each client. This prevents the AS from accessing the data held by each individual client.

In summary, our homomorphic encryption scheme remains semantically secure as long as fewer than z clients are involved in a conspiracy with the AS.

B. Convergence analysis

In this section, we analyze the convergence properties of our proposed Algorithm 1. We first state a list of assumptions below.

Assumption 1: The global model converges with rate $g(t)$, where

$$\lim_{t\to\infty} g(t) = 0, ||w^t - w^*||_2 \leq g(t) \tag{13}$$

Assumption 2: The expectation of stochastic gradients is uniformly bounded across all clients and iterations.

$$||\nabla F_i(w)||_2 \leq Q \tag{14}$$

Assumption 3: The L2 distance between the optimal local models and the optimal global model is bounded:

$$||w_i^* - w^*||_2 \leq M \tag{15}$$

Lemma 1 : Based on the above assumptions and properties, in each round of communication, client c_i is selected with a local step size decay of $\eta = \frac{2}{(t+1)(\varepsilon+\tau)}$. Consequently, in each round of communication, we obtain the following:

$$\begin{aligned}
&||w_i^{(t+1)} - w_i^*||_2 \\
&\leq \left(1 - \frac{2}{t+1}\right)\frac{Q^2}{\varepsilon^2} + \frac{4}{(t+1)^2\left(\varepsilon + \frac{E}{2\sigma}\right)} \\
&+ \left(Q^2 + \left(\frac{E}{2\sigma}\right)^2 + \frac{E}{\sigma}\left(\frac{Q}{\varepsilon} + M\right)\right) \\
&+ \left(\frac{\varepsilon E}{2\sigma}\right)^2 ||w^t - w^*||_2 + \frac{8}{(t+1)^2\left(\varepsilon + \frac{E}{2\sigma}\right)^2} \\
&\quad \left(Q + \frac{E}{2\sigma}\left(M + \frac{Q}{\varepsilon}\right)\right)||w^t - w^*|| \\
&+ \frac{\varepsilon E}{\sigma}||v_i^t - v_i^*||\,||w^t - w^*||
\end{aligned} \tag{16}$$

Proof : Denote $g(w_i^t; w^t)$ as the stochastic gradient of $\mathcal{L}_\rangle(w_i^t; w^t)$.

$$\begin{aligned}
&||w_i^{(t+1)} - w_i^*||_2 = ||w_i^t - \eta g(w_i^t; w^t) - w_i^*||_2 \\
&= ||w_i^t - w_i^*||_2 + \eta^2||g(w_i^t; w^t)||_2 - 2\eta\left\langle g(w_i^t), w_i^t - w_i^*\right\rangle \\
&\leq (1 - (\varepsilon + \frac{E}{2\sigma})\eta)||w_i^t - w_i^*||_2 \\
&\quad + \eta^2||g(w_i^t; w^t)||_2 - 2\eta||\mathcal{L}_i(w_i^t; w^t) - \mathcal{L}_i(w_i^*; w^t)|| \\
&\leq (1 - (\varepsilon + \frac{E}{2\sigma}))||w_i^t - w_i^*||_2 \\
&\quad + \eta^2||g(w_i^t; w^*)||_2 - (\eta\frac{E}{2\sigma})^2||w^t - w^*||_2 \\
&\quad - 2\eta^2\frac{E}{2\sigma}||g(w_i^t; w^*)||||w^t - w_*|| - 2\eta(\mathcal{L}_i(w_i^t; w^t) \\
&\quad - \mathcal{L}_i(w_i^*; w^t))) + 2\eta\frac{E}{2\sigma}||w_i^t - w_i^*||||w^t - w^*||
\end{aligned} \tag{17}$$

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It follows from the properties of strongly convex functions:

$$\varepsilon^2||w_i^t - w_i^*||_2 \leq ||\nabla F_i(w_i^t)||_2$$

$$||w_i^t - w_i^*||_2 \leq \frac{1}{\varepsilon^2}||\nabla F_i(w_i^t)||_2 \leq \frac{Q^2}{\varepsilon^2} \tag{18}$$

$$\begin{aligned} ||w_i^t - w^*|| &= ||w_i^t - w_i^* + w_i^* - w^*||_2 \\ &\leq \frac{Q^2}{\varepsilon^2} + M^2 + \frac{2MQ}{\varepsilon} \end{aligned} \tag{19}$$

$$\begin{aligned} ||g(w_i^t; w^*) &= ||\nabla F_i(w_i^t) + r||w_i^t - w^*||_2 \\ &\leq Q^2 + \left(\frac{E}{2\sigma}\right)^2 \left(\frac{Q}{\varepsilon} + M\right)^2 + \\ &\quad 2Q\frac{E}{2\sigma}\left(\frac{Q}{\varepsilon} + M\right) \end{aligned} \tag{20}$$

Substituting Eq. 18, Eq. 19, and Eq. 20 into Eq. 17 yields Eq.16. □

Lemma 2 : Under the assumptions above, when the global objective $G(\cdot)$ is FedAvg, $\eta = \frac{2}{\varepsilon(t+1)}$ then under Algorithm 1, for $i \in [K]$,

$$F_i(w_i^t) - F_i(w_i^*) = O(\frac{1}{n}) \tag{21}$$

Proof :

$$\begin{aligned} ||w_i^{t+1} - w_i^*||_2 &= ||w^t - w^*||_2 - 2\eta \left\langle g^t(w_i^t; w^*), w_i^t - w_i^* \right\rangle \\ &\quad + \eta^2||g^t(w_i^t; w^*)||_2 \\ &\leq (1 - \varepsilon\eta)||w^t - w^*||_2 \\ &\quad - 2\eta\left(F_i(w_i^k) - F_i(w_i^*)\right) \\ &\quad + \varepsilon^2||g^t(w_i^t; w^*)||_2 \end{aligned} \tag{22}$$

$$\begin{aligned} F_i(w_i^t) - F_i(w_i^*) &\leq \frac{\varepsilon(t-1)}{4}||w^t - w^*||_2 \\ &\quad + \frac{1}{\varepsilon(t+1)}||g^t(w_i^t; w^*)||_2 - \frac{\varepsilon(t+1)}{4}||w_i^{t+1} - w_i^*||_2 \end{aligned} \tag{23}$$

Bringing $\eta = \frac{2}{\varepsilon(t+1)}$ into the Eq.23 yields:

$$\begin{aligned} F_i(w_i^t) - F_i(w_i^*) &\leq \frac{\varepsilon(t-1)}{4}||w^t - w^*||_2 \\ &\quad + \frac{1}{\varepsilon(t+1)}||g^t(w_i^t; w^*)||_2 - \frac{\varepsilon(t+1)}{4}||w_i^{t+1} - w_i^*||_2 \end{aligned} \tag{24}$$

Multiplying the left and right sides of the inequality by t yields:

$$\sum_{t=0}^{n} t(F_i(w_i^t) - F_i(w_i^*)) \leq 0 + \sum_{t=0}^{n} \frac{t}{\varepsilon(t+1)}||g^t(w_i^t; w^*)||_2$$

$$- \frac{\varepsilon n(n+1)}{4}||w_i^{n+1} - w_i^*||_2 \leq \frac{n||g^t(w_i^t; w^*)||_2}{\varepsilon} \tag{25}$$

$$F_i(w_i^t) - F_i(w_i^*) \leq \frac{2||g^t(w_i^t; w^*)||_2}{\varepsilon(n+1)} \tag{26}$$

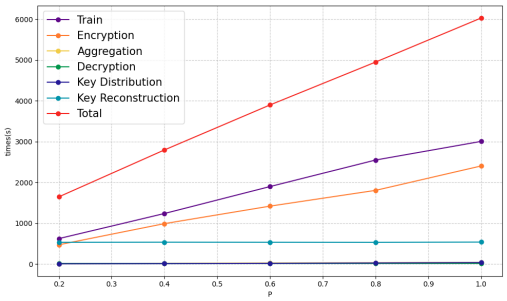
as desired. □

VI. EXPERIMENTAL EVALUATION

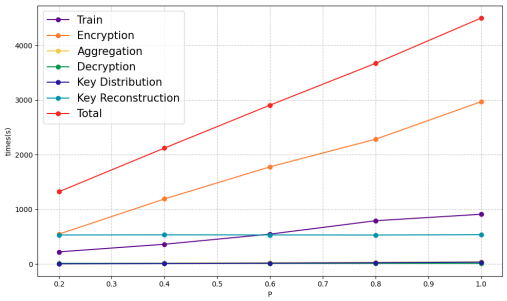
In this section, we evaluate our method within the context of various dataset distributions. We show the efficiency, convergence, and accuracy of our method's execution.

Our experiments utilized the HAR (Human Activity Recognition) dataset [23], which includes data from 30 individuals engaged in six activities: walking, walking upstairs, walking downstairs, sitting, standing, and lying down. The dataset consists of 10,299 instances and 561 features, with each individual contributing approximately 300 instances. We allocated 75% of the data for training the model and reserved 25% for testing purposes.

Our experimental setup involved a personal computer equipped with a 2.1 GHz Intel Core i7 processor, running a 64-bit Windows 10 operating system, and having 16.00 GB of RAM.



(a) Execution time (batch size=10)



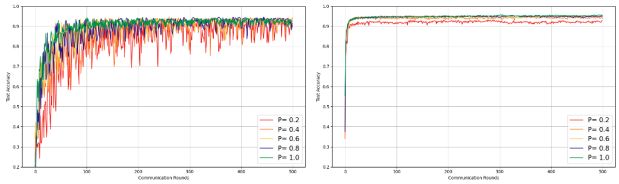
(b) Execution time (batch size=40)

Fig. 3: Execution times under different batch sizes

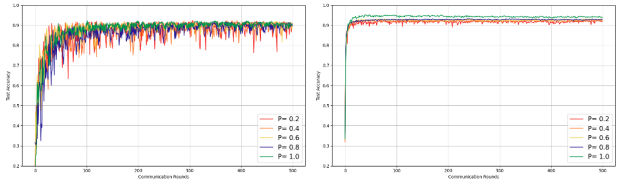
In a federated setting, several key factors can impact the performance of federated learning: (1) the proportion of clients participating in each round (P), (2) the local batch size, and (3) the distribution of the dataset (IID). We begin by conducting experiments on the proportion of clients participating in each round of communication. First, we set $IID = NO$ and $IID = YES$, separately and then evaluate the test accuracy at different values of P and batch size. More specifically, we investigate the effect of P and batch size by setting them to $0.2, 0.4, 0.6, 0.8, 1$, and $10, 40$.

As shown in Fig.3(a) and Fig.3(b), we analyze the computational overhead of different parameters. It can be observed

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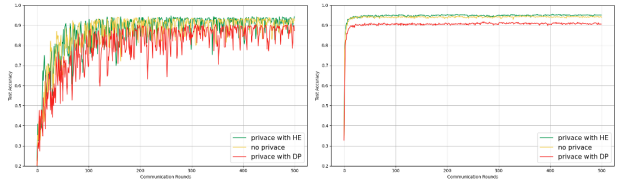


(a) non-IID dataset (batch size=10) (b) IID dataset (batch size=10)



(c) non-IID dataset (batch size=40) (d) IID dataset (batch size=40)

Fig. 4: Test accuracy under different number of participating clients in each communication round (batch size=10 and batch size=40)



(a) non-IID dataset (b) IID dataset

Fig. 5: Test accuracy of different FL approaches under HAR.

that from Fig.3(a) when $batchsize = 10$ the encryption time and the model training time on the client side constitute the majority of the total time overhead. For example, when $P = 0.4$, encryption accounts for 35.37% of the total time, while the client-side model training accounts for 44.23%, and as depicted in Fig.3(b), we find that when $batchsize = 40$, the encryption time and key reconstruction time for smaller values of P account for most of the total time overhead. For example, when $P = 0.4$, encryption accounts for 56.09% of the total time, while key reconstruction accounts for 25.15%. For larger values of P, the client's model training time and encryption time dominate the total time overhead. For instance, when $P = 0.8$, model training accounts for 21.57% of the total time, and encryption accounts for 62.18%.

As depicted in Fig.4, the learning algorithms converge within 500 communication rounds across different choices of P and B in the IID dataset. Additionally, with the increasing of P, the convergence in the non-IID dataset also gradually occurs within 500 communication rounds.

The test accuracy improve with larger values of P and more communication rounds. When P is small, there is a noticeable gap in test accuracy compared to larger P. For example, with B=10 in the non-IID dataset, the test accuracy is 90.83% when $P = 0.2$ and 93.33% when $P = 1$. In the IID dataset under the same conditions, the test accuracy is 91.30% for $P = 0.2$ and 95.26% for $P = 1$. When $batchsize = 40$ in the non-IID dataset, the test accuracy is 91.15% with $P = 0.2$ and 95.23% with $P = 1$. In the IID dataset, the test accuracy is 93.15% for $P = 0.2$ and 96.23% for $P = 1$.

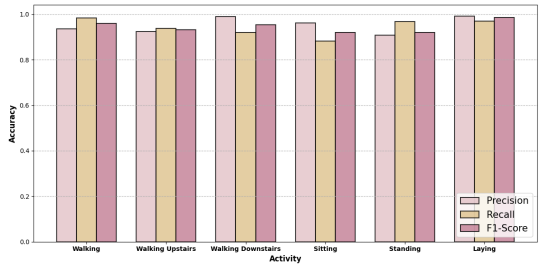


Fig. 6: Average Precision, Recall, and F1-Score for Each Activity

Considering the robustness of the system, as well as computation time and accuracy, we choose $batchsize = 40$ and $P = 0.6$ for the subsequent experiments. We compare our scheme with a scheme that uses traditional objective functions without privacy preservation and a scheme using differential privacy[3] as shown in Fig. 5, since utilizing differential privacy methods for privacy preservation introduces noise into the parameters, it leads to a reduction in effectiveness. However, our scheme uses encryption to protect the parameters, avoiding the introduction of additional noise. Additionally, our new objective function improves the model's performance, yielding an approximate 1% increase in accuracy over the model using the traditional objective function and about a 4% improvement over the model employing differential privacy preservation.

To give a more detailed performance evaluation, when set $batchsize = 40$ and $P = 0.6$ we also show the average precision, recall and f1-score for each activity. As shown in Fig. 6, our scheme is able to achieve excellent performance on the metrics of all activities. Notably, the accuracy for the Laying activity exceeds 99%, demonstrating the scheme's effectiveness in identifying falling actions.

VII. CONCLUSION

In this paper, we introduce a homomorphic encryption-based privacy-preserving federated learning (FL) framework designed to ensure the privacy and security of client data storage. Our scheme can aggregates models from various clients to generate a global model without exposing the model parameters. It enhances model training by optimizing the gradient between local and global models. Experimental results on real datasets demonstrate that our framework achieves high classification accuracy and privacy preservation with low computational and communication overhead. These

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security, privacy, and communication properties make our approach applicable across various domains, such as healthcare, education, banking, and other areas where data security is essential.

REFERENCES

[1] M. Soori, B. Arezoo, and R. Dastres, "Internet of things for smart factories in industry 4.0, a review," *Internet of Things and Cyber-Physical Systems*, 2023.

[2] U. Ahmed, I. Raza, and S. A. Hussain, "Trust evaluation in cross-cloud federation: Survey and requirement analysis," *ACM Computing Surveys (CSUR)*, vol. 52, no. 1, pp. 1–37, 2019.

[3] R. Hu, Y. Guo, H. Li, Q. Pei, and Y. Gong, "Personalized federated learning with differential privacy," *IEEE Internet of Things Journal*, vol. 7, no. 10, pp. 9530–9539, 2020.

[4] X. Wu, Y. Zhang, M. Shi, P. Li, R. Li, and N. N. Xiong, "An adaptive federated learning scheme with differential privacy preserving," *Future Generation Computer Systems*, vol. 127, pp. 362–372, 2022.

[5] K. Wei, J. Li, C. Ma, *et al.*, "Personalized federated learning with differential privacy and convergence guarantee," *IEEE Transactions on Information Forensics and Security*, 2023.

[6] A. Acar, H. Aksu, A. S. Uluagac, and M. Conti, "A survey on homomorphic encryption schemes: Theory and implementation," *ACM Computing Surveys (Csur)*, vol. 51, no. 4, pp. 1–35, 2018.

[7] N. Koblitz, A. Menezes, and S. Vanstone, "The state of elliptic curve cryptography," *Designs, codes and cryptography*, vol. 19, pp. 173–193, 2000.

[8] T. Li, S. Hu, A. Beirami, and V. Smith, "Ditto: Fair and robust federated learning through personalization," in *International conference on machine learning*, PMLR, 2021, pp. 6357–6368.

[9] S. P. Karimireddy, S. Kale, M. Mohri, S. Reddi, S. Stich, and A. T. Suresh, "Scaffold: Stochastic controlled averaging for federated learning," in *International conference on machine learning*, PMLR, 2020, pp. 5132–5143.

[10] A. Bhowmick, J. Duchi, J. Freudiger, G. Kapoor, and R. Rogers, "Protection against reconstruction and its applications in private federated learning," *arXiv preprint arXiv:1812.00984*, 2018.

[11] M. Nasr, R. Shokri, and A. Houmansadr, "Comprehensive privacy analysis of deep learning: Passive and active white-box inference attacks against centralized and federated learning," in *2019 IEEE symposium on security and privacy (SP)*, IEEE, 2019, pp. 739–753.

[12] C. Dwork, "Differential privacy," in *International colloquium on automata, languages, and programming*, Springer, 2006, pp. 1–12.

[13] Q. Wu, X. Chen, Z. Zhou, and J. Zhang, "Fedhome: Cloud-edge based personalized federated learning for in-home health monitoring," *IEEE Transactions on Mobile Computing*, vol. 21, no. 8, pp. 2818–2832, 2020.

[14] B. Wang, H. Li, Y. Guo, and J. Wang, "Ppflhe: A privacy-preserving federated learning scheme with homomorphic encryption for healthcare data," *Applied Soft Computing*, vol. 146, p. 110 677, 2023.

[15] L. Zhang, J. Xu, P. Vijayakumar, P. K. Sharma, and U. Ghosh, "Homomorphic encryption-based privacy-preserving federated learning in iot-enabled healthcare system," *IEEE Transactions on Network Science and Engineering*, 2022.

[16] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Artificial intelligence and statistics*, PMLR, 2017, pp. 1273–1282.

[17] C. Marcolla, V. Sucasas, M. Manzano, R. Bassoli, F. H. Fitzek, and N. Aaraj, "Survey on fully homomorphic encryption, theory, and applications," *Proceedings of the IEEE*, vol. 110, no. 10, pp. 1572–1609, 2022.

[18] J. H. Cheon, A. Kim, M. Kim, and Y. Song, "Homomorphic encryption for arithmetic of approximate numbers," in *Advances in Cryptology–ASIACRYPT 2017: 23rd International Conference on the Theory and Applications of Cryptology and Information Security, Hong Kong, China, December 3-7, 2017, Proceedings, Part I 23*, Springer, 2017, pp. 409–437.

[19] M. R. Albrecht, R. Player, and S. Scott, "On the concrete hardness of learning with errors," *Journal of Mathematical Cryptology*, vol. 9, no. 3, pp. 169–203, 2015.

[20] H. Bennett, "The complexity of the shortest vector problem," *ACM SIGACT News*, vol. 54, no. 1, pp. 37–61, 2023.

[21] D. Micciancio, "The hardness of the closest vector problem with preprocessing," *IEEE Transactions on Information Theory*, vol. 47, no. 3, pp. 1212–1215, 2001.

[22] H. K. Maji, H. H. Nguyen, A. Paskin-Cherniavsky, T. Suad, and M. Wang, "Leakage-resilience of the shamir secret-sharing scheme against physical-bit leakages," in *Annual International Conference on the Theory and Applications of Cryptographic Techniques*, Springer, 2021, pp. 344–374.

[23] D. Anguita, A. Ghio, L. Oneto, X. Parra, J. L. Reyes-Ortiz, *et al.*, "A public domain dataset for human activity recognition using smartphones.," in *Esann*, vol. 3, 2013, p. 3.

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